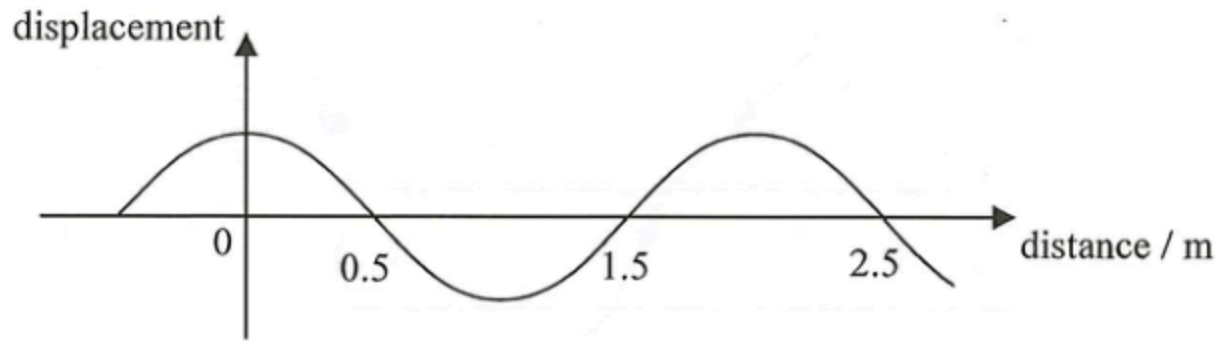


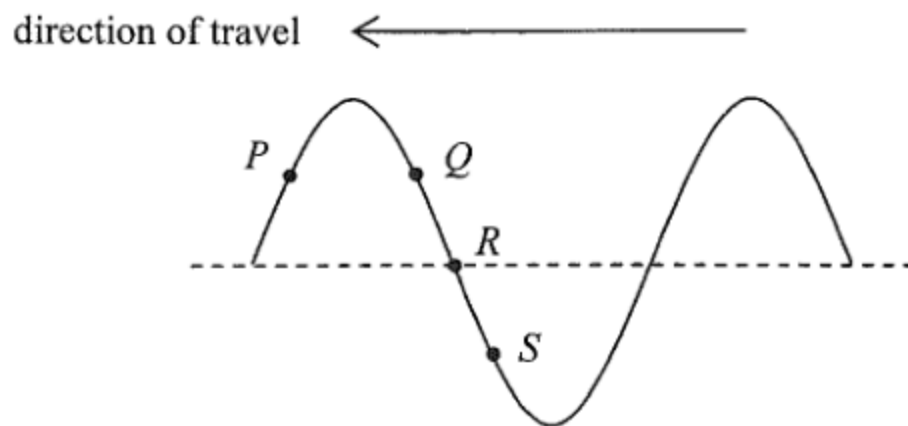
14.



The above figure shows the displacement-distance graph of a wave travelling at a speed of 10 m s^{-1} . What is the period of this wave?

- A. 0.10 s
- B. 0.15 s
- C. 0.20 s
- D. 0.25 s

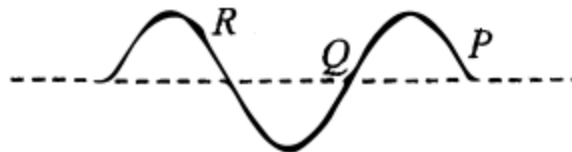
14. A transverse wave travels towards the left on a long string. P , Q , R and S are particles on the string. Which of the following statements correctly describe(s) their motions at the instant shown ?



- (1) P is moving upwards.
- (2) Q and S are moving in opposite directions.
- (3) R is momentarily at rest.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

15.



The above figure shows a snapshot of a transverse wave which travels along a string. Which statement is correct ?

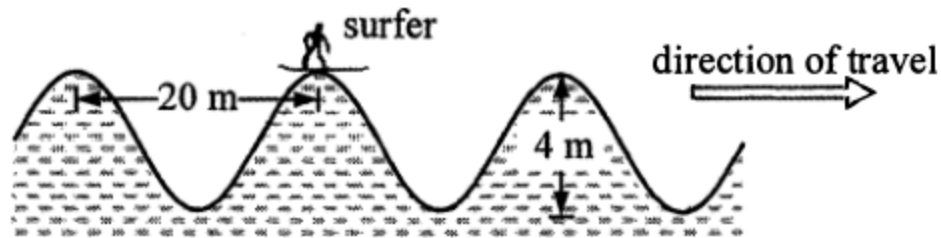
- A. The wave is travelling to the left if particle P is moving upwards at this instant. ✓
- B. Particles P and R are moving in the same direction at this instant.
- C. Particle Q is at rest at this instant.
- D. Particle R vibrates with an amplitude larger than that of particle Q .

10. Which of the following can be transferred by mechanical waves from one place to another along the direction of propagation ?

- (1) mass
- (2) momentum
- (3) energy

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

16.



The surfer in the figure reaches a crest at the moment shown. The crests of the water wave are 20 m apart and the surfer descends a vertical distance of 4 m from a crest to a trough in a time interval of 2 s. What is the speed of the wave ?

- A. 1 m s^{-1}
- B. 2 m s^{-1}
- C. 5 m s^{-1}
- D. 10 m s^{-1}

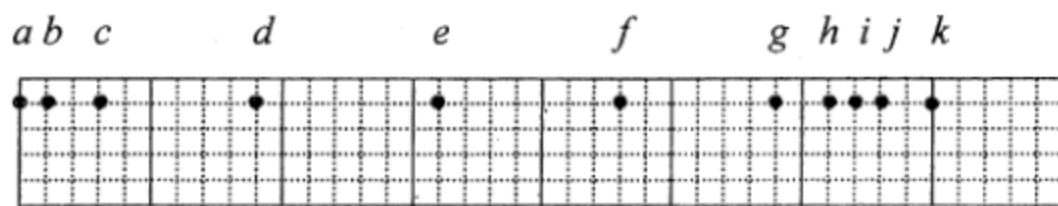


Figure (a)

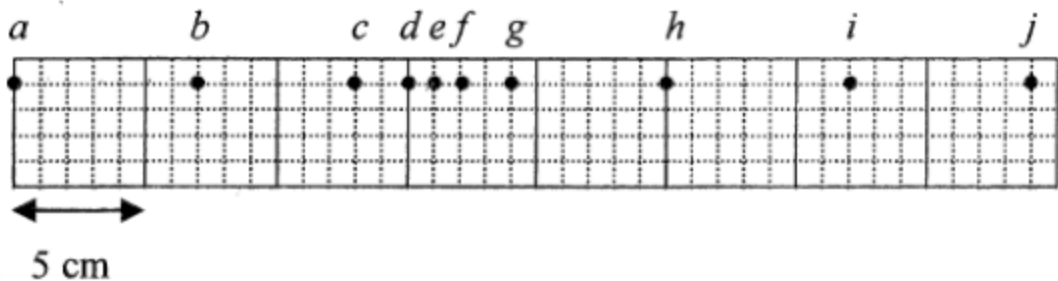


Figure (b)

A series of particles is uniformly distributed along a slinky spring initially. Figure (a) shows their positions at a certain instant when a travelling wave propagates along the slinky spring from left to right. Figure (b) shows their positions 0.1 s later. Which statement is correct ?

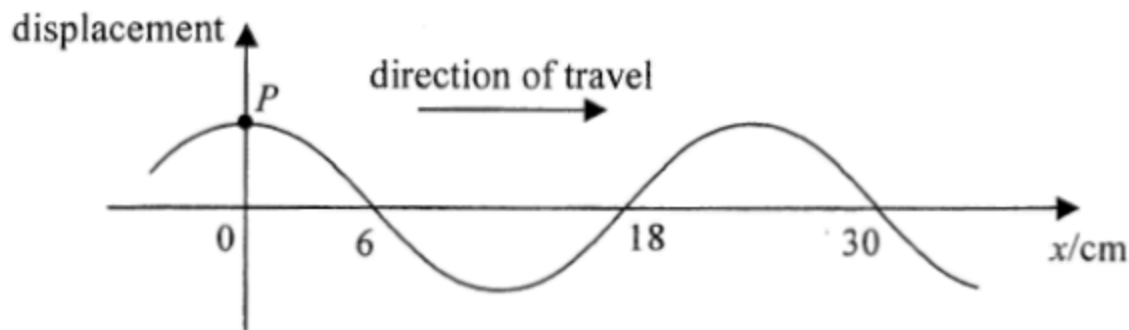
- A. Particle *e* is always stationary.
- B. Particles *a* and *i* are in phase.
- C. The wavelength of the wave is 16 cm.
- D. The frequency of the wave is 10 Hz.

14. Which of the following statements about waves is/are correct ?

- (1) Longitudinal waves can transmit energy from one place to another but transverse waves cannot.
- (2) Sound waves propagate faster in water than in air.
- (3) Infra-red radiation is a kind of electromagnetic wave.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

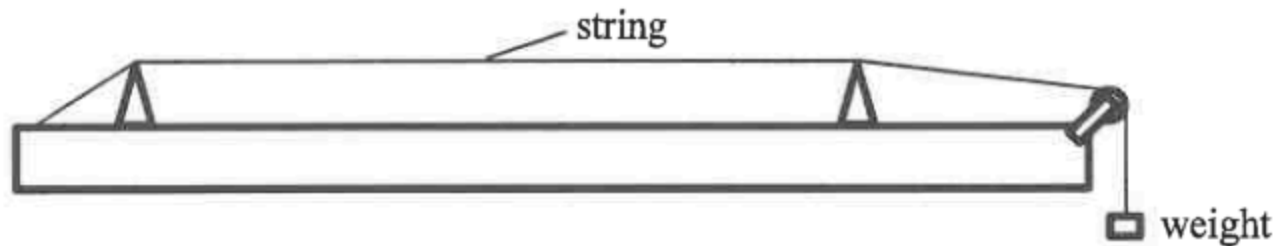
16.



The figure shows a snapshot of a section of a continuous transverse wave travelling along the x -direction at time $t = 0$. At $t = 1.5$ s, the particle P just passes the equilibrium position for a *second* time at that moment. Find the wave speed.

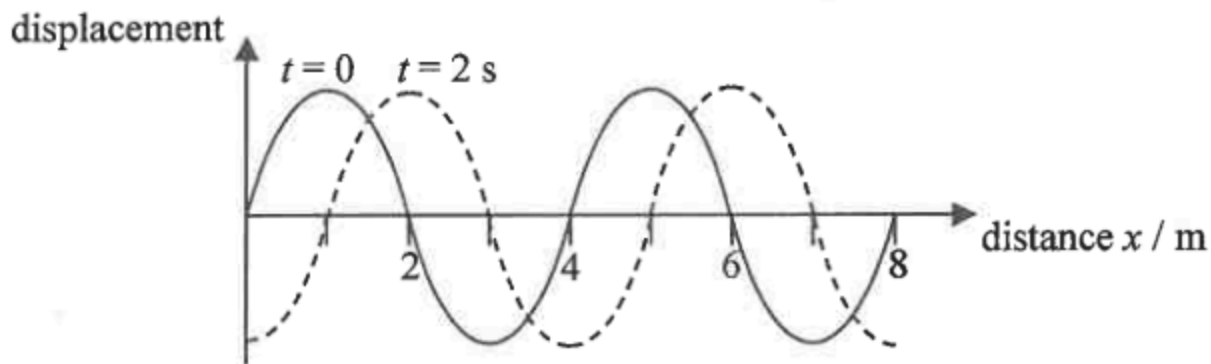
- A. 20 cm s^{-1}
- B. 12 cm s^{-1}
- C. 6 cm s^{-1}
- D. 4 cm s^{-1}

16. Referring to the set-up below, which of the following combinations would give the highest speed of wave propagation along the string when the string is plucked ?



	tension of the string	radius of the cross-section of the string
A.	T	r
B.	T	$2r$
C.	$2T$	r
D.	$2T$	$2r$

15. The figure shows the displacement-distance graph at time $t = 0$ and $t = 2$ s respectively of a wave travelling to the right.



Which statements about the wave are correct ?

- (1) The wavelength is 4 m.
- (2) The period is 4 s.
- (3) The speed of the wave can be 2.5 m s^{-1} .

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

18. How does the speed of propagation of waves along a stretched string change if the tension in the string is increased or the string is replaced by a more massive one of the same length and tension ?

tension increased

**using a more massive string of
the same length and tension**

- A. speed increases
- B. speed increases
- C. speed decreases
- D. speed decreases

- speed decreases
- speed increases
- speed decreases
- speed increases

12. Earthquakes propagate in the form of waves. The quake centre produces both longitudinal waves (P-wave) and transverse waves (S-wave) which travel with speeds 9.6 km s^{-1} and 6.4 km s^{-1} respectively on the Earth's crust. In an earthquake, a monitoring station detects the arrival of the P-wave pulse at 7:02 a.m. while the S-wave pulse arrives 2 minutes later at 7:04 a.m. Estimate the time that this earthquake occurs.

- A. 6:53 a.m.
- B. 6:56 a.m.
- C. 6:58 a.m.
- D. 6:59 a.m.

17.

Figure (1)

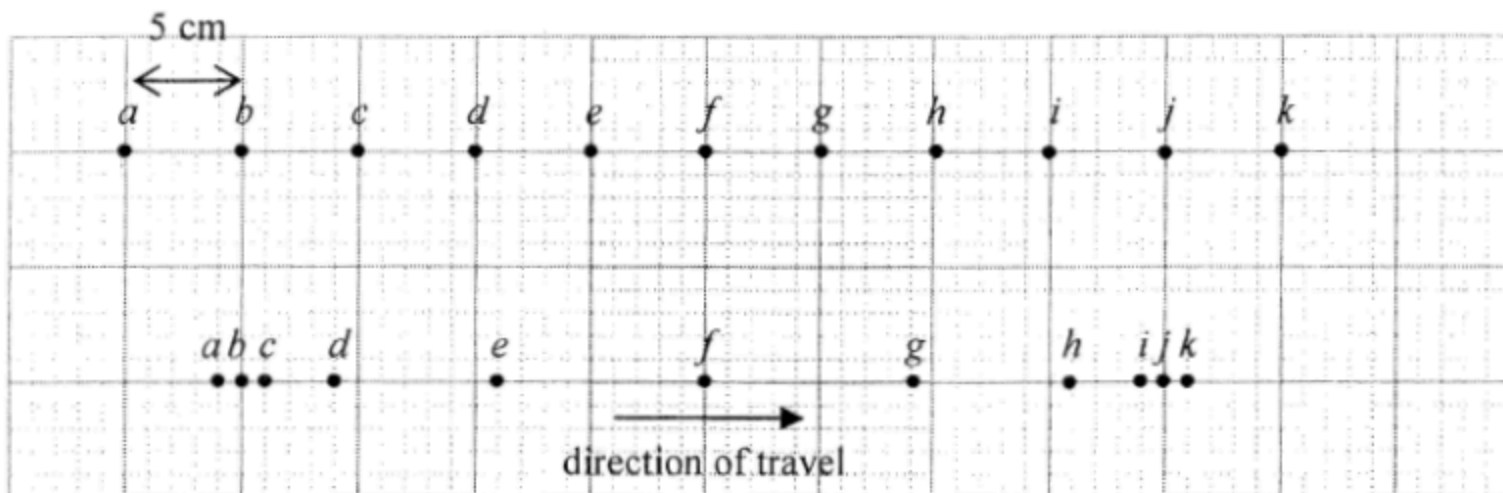


Figure (1) shows the equilibrium positions of particles *a* to *k* separated by 5 cm from each other in a medium. A longitudinal wave is travelling from left to right with a speed of 80 cm s^{-1} . At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and frequency of the wave.

amplitude frequency

- | | | |
|----|------|------|
| A. | 6 cm | 2 Hz |
| B. | 6 cm | 4 Hz |
| C. | 9 cm | 2 Hz |
| D. | 9 cm | 4 Hz |

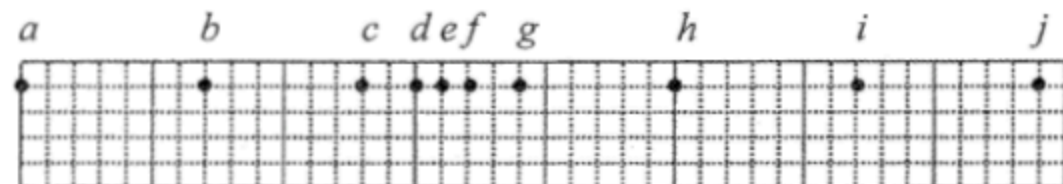


Figure (a)

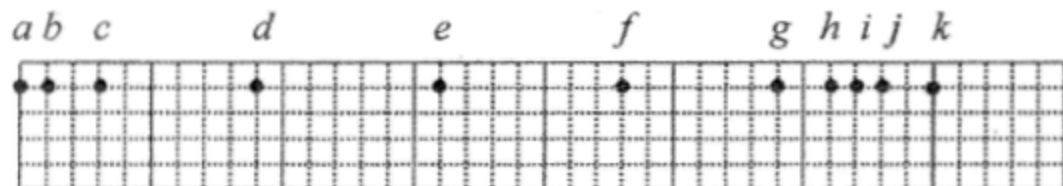


Figure (b)

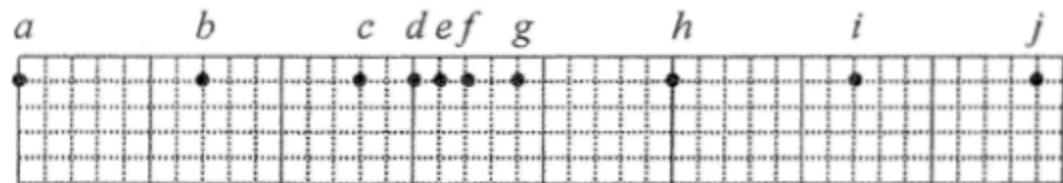


Figure (c)

A series of particles is uniformly distributed along a slinky spring initially. When a travelling wave propagates along the slinky spring from left to right, Figure (a) shows the positions of the particles at a certain instant. Figures (b) and (c) respectively show their positions 0.05 s and 0.1 s later. Which of the following is/are a possible frequency of the wave ?

- (1) 10 Hz
- (2) 20 Hz
- (3) 40 Hz

- A. (1) only
- B. (2) only
- C. (3) only
- D. (1), (2) and (3)

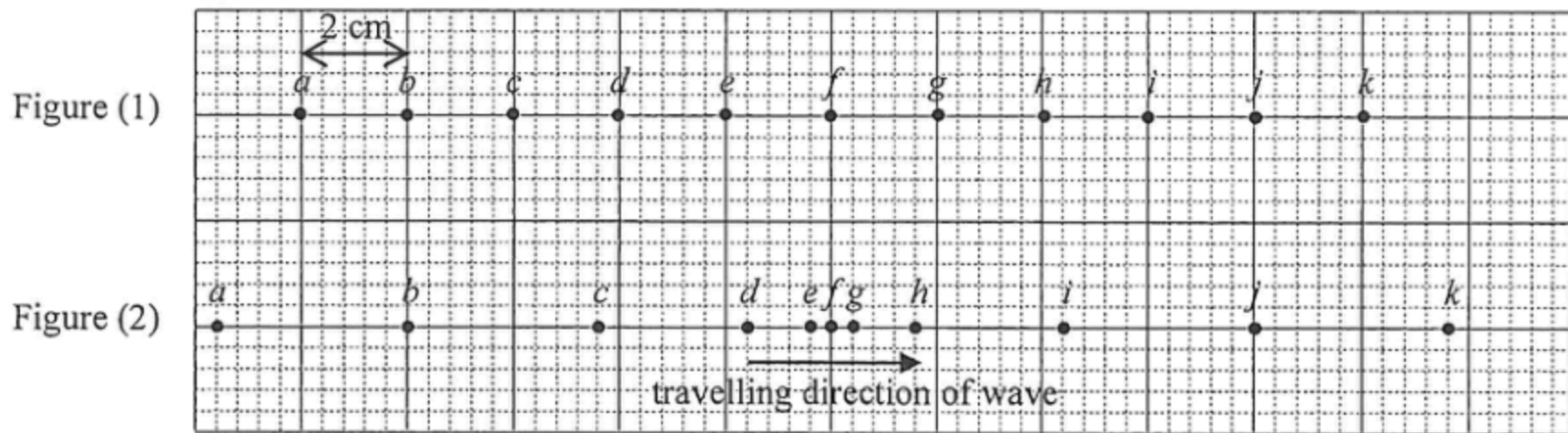
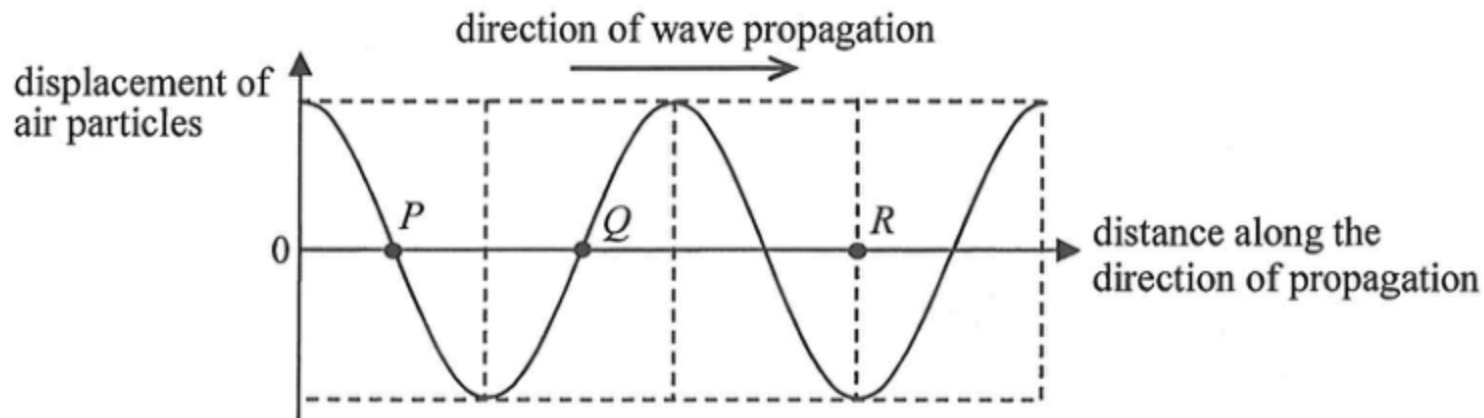


Figure (1) shows the equilibrium positions of particles *a* to *k* in a medium. The particles are separated by 2 cm from each other. A longitudinal wave of frequency 5 Hz is travelling from left to right. At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and speed of the wave.

	amplitude	speed
A.	3.6 cm	40 cm s^{-1}
B.	3.6 cm	80 cm s^{-1}
C.	2.4 cm	40 cm s^{-1}
D.	2.4 cm	80 cm s^{-1}

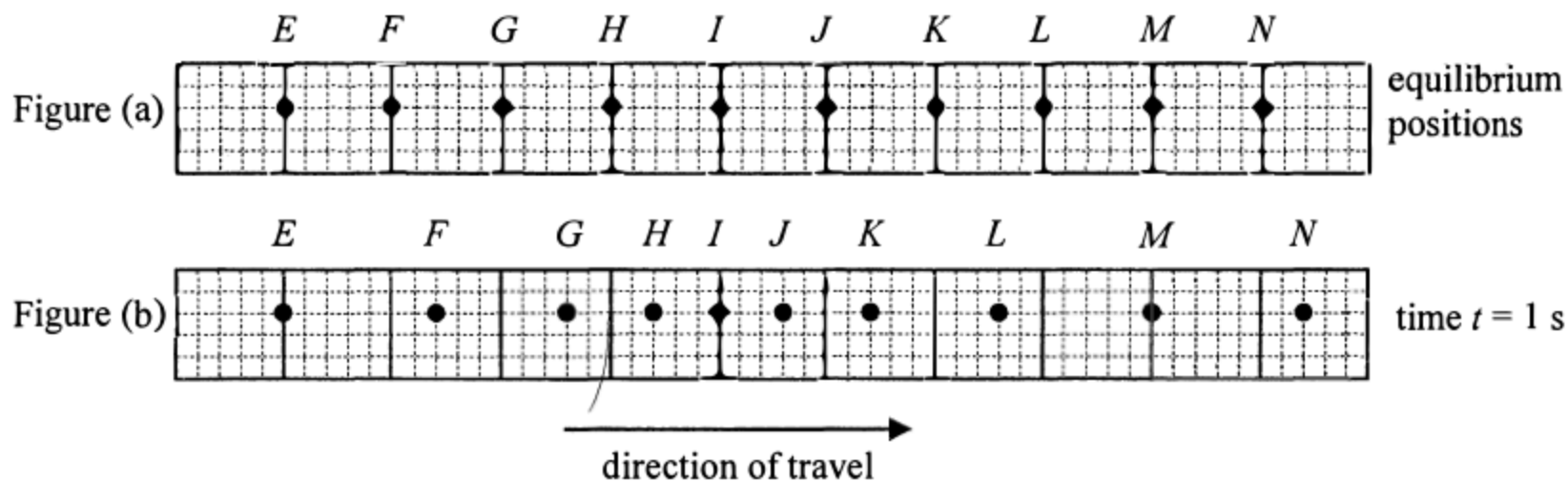
15. The graph shows the variation of the displacement of air particles from their equilibrium positions with distance for a sound wave at a certain instant. Displacement along the direction of wave propagation is taken to be positive.



Which statement concerning the wave at that instant is correct ?

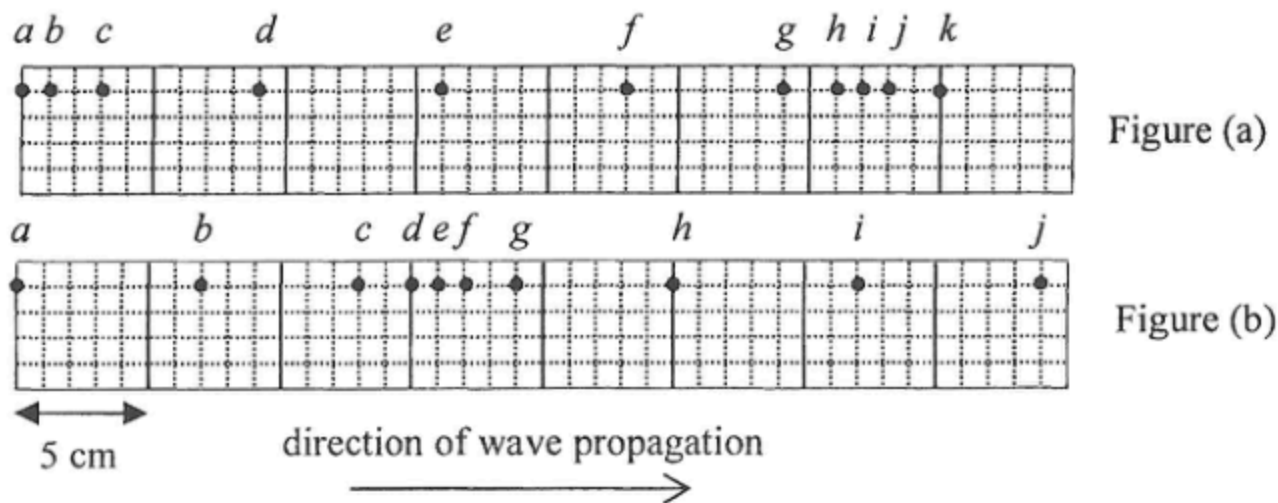
- A. An air particle at P is momentarily at rest.
- B. A rarefaction occurs at Q .
- C. A compression occurs at R .
- D. An air particle at R is moving with the greatest speed.

14. Figure (a) shows the equilibrium positions of particles E to N in a medium. At time $t = 0$, a longitudinal wave starts travelling from left to right. At time $t = 1$ s, the positions of the particles are shown in Figure (b).



Which of the following statements **MUST BE** correct ?

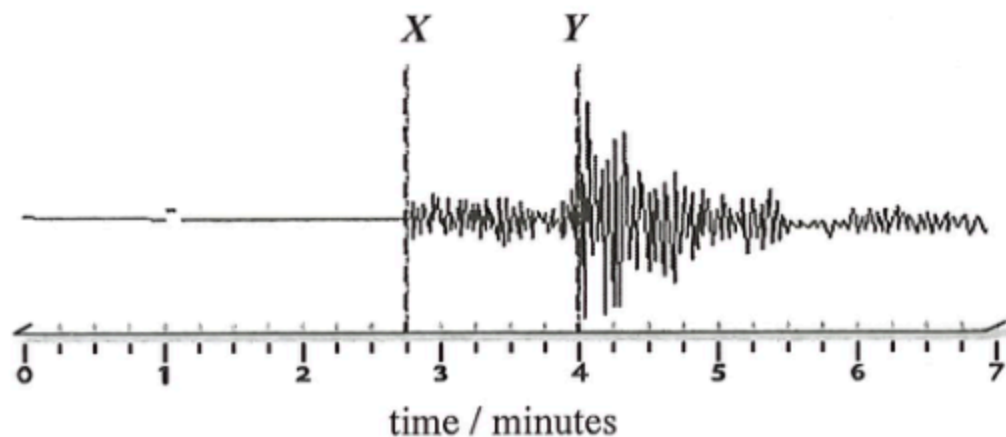
- A. The distance between particles F and N is equal to the wavelength of the wave.
- B. The period of the wave is 1 s.
- C. Particle E is always at rest.
- D. Particle I is momentarily at rest at $t = 1$ s.



A series of particles is uniformly distributed along a slinky spring initially. Figure (a) and (b) show their positions at two instants when a travelling wave of frequency 10 Hz propagates along the spring from left to right. Which statement below is correct ?

- A. Particle *a* is always stationary.
- B. Particle *b* and *c* vibrate with different amplitudes.
- C. The wavelength of the wave is 16 cm.
- D. The smallest time interval for obtaining these two figures is 0.05 s.

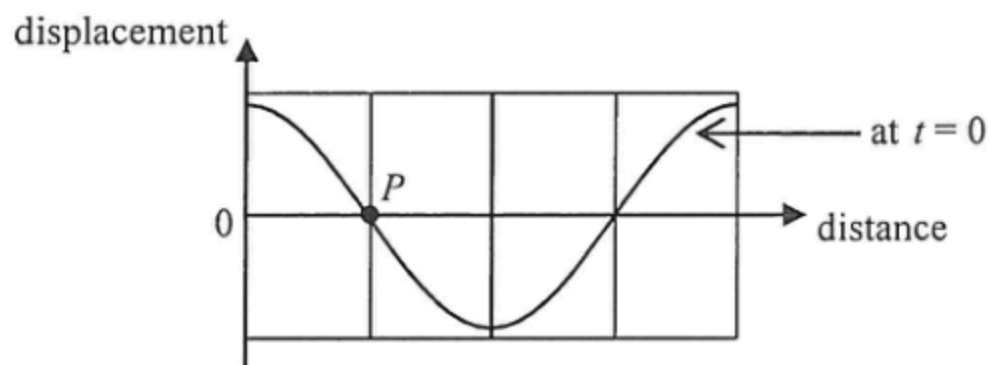
15. Earthquakes produce P-waves (longitudinal) and S-waves (transverse) which travel at 7.0 km s^{-1} and 4.0 km s^{-1} respectively. On a given day, a seismometer detected the P- and S- waves from a shallow earthquake (see the figure below) with its quake centre located at a distance D from the seismometer.



Which deduction below is correct ?

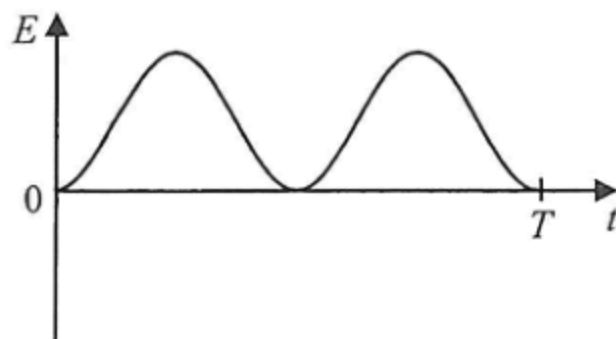
	Y in the figure denotes the arrival of	D / km
A.	P-waves	225
B.	P-waves	700
C.	S-waves	225
D.	S-waves	700

12. The figure shows part of the displacement-distance graph of a travelling wave of period T at time $t = 0$. P is a particle on the wave.

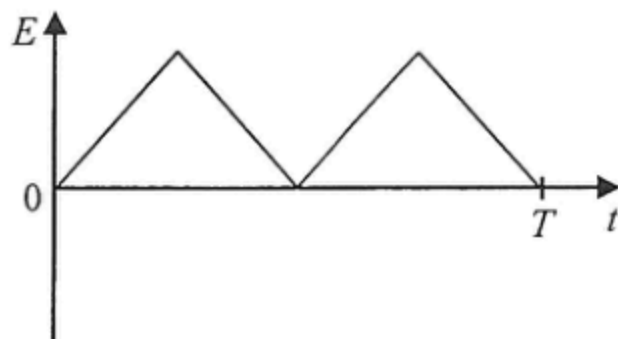


Which graph below correctly shows the variation of the particle's kinetic energy E within a period starting from $t = 0$?

A.



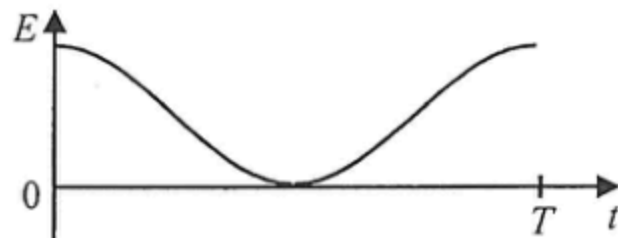
B.



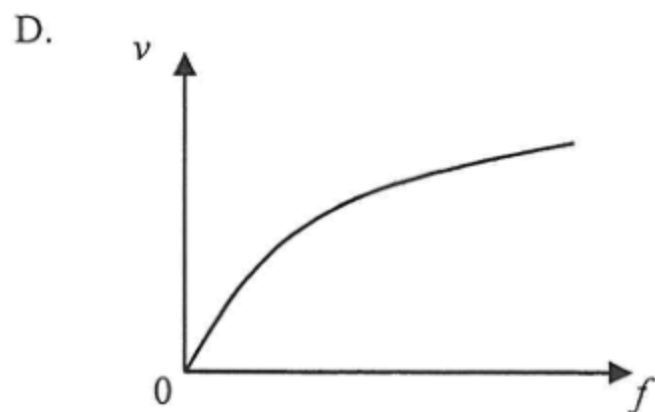
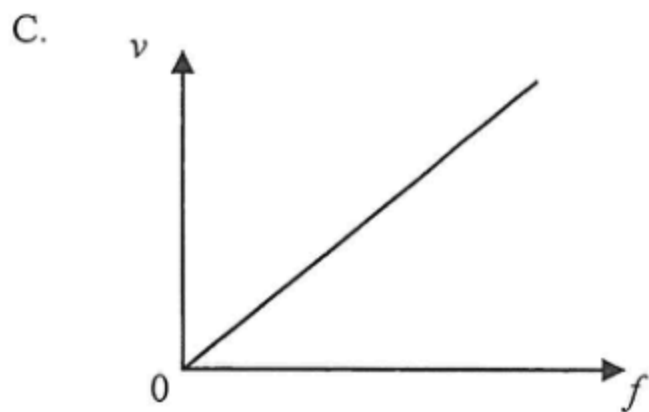
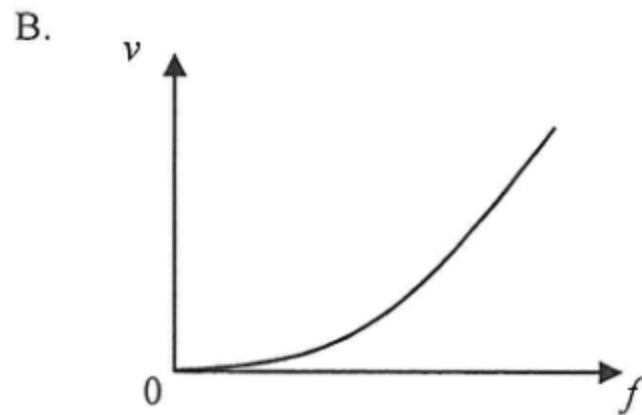
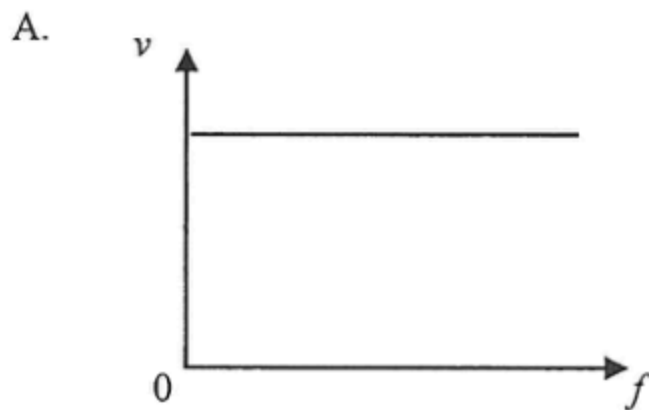
C.



D.

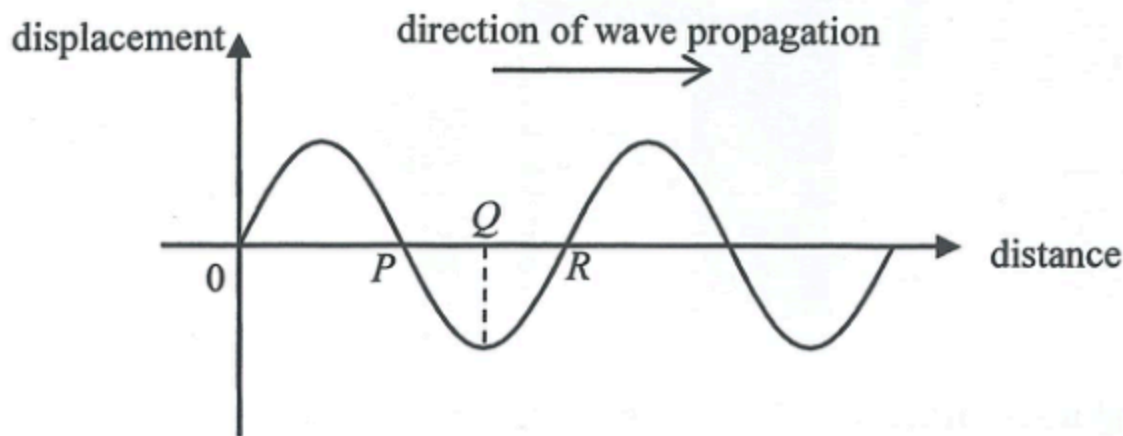


16. A transverse wave propagates along a stretched string. Which graph below correctly shows the variation of the speed v of the wave with its frequency f ?



14.

The figure shows the displacement-distance graph at a certain instant of a longitudinal wave which travels to the right. Displacement to the right is taken to be positive.

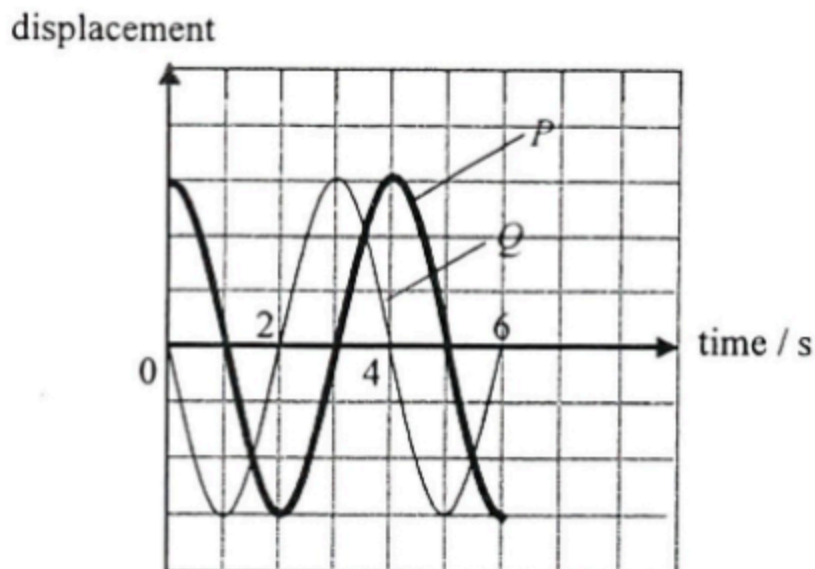


At the instant shown, which of the following statements is/are correct ?

- (1) P is a centre of compression.
- (2) A particle with its equilibrium position at Q is at rest.
- (3) A particle with its equilibrium position at R is moving downwards.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

15. The figure below shows the displacement-time graph of particles P and Q on the same transverse travelling wave of wavelength λ .

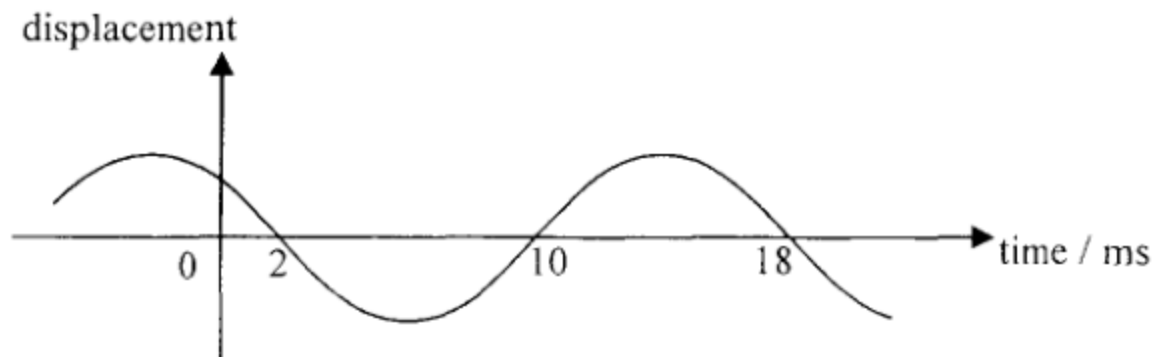


Which of the following statements **MUST BE** correct? Upward displacement is taken to be positive.

- (1) At time $t = 2$ s, P is momentarily at rest.
- (2) At time $t = 4$ s, Q is moving downwards.
- (3) The separation between the equilibrium positions of P and Q is 0.25λ .

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

11.



The displacement-time graph of a particle on a travelling wave is as shown. Find the frequency of the wave.

- A. 55.6 Hz
- B. 62.5 Hz
- C. 111 Hz
- D. 125 Hz

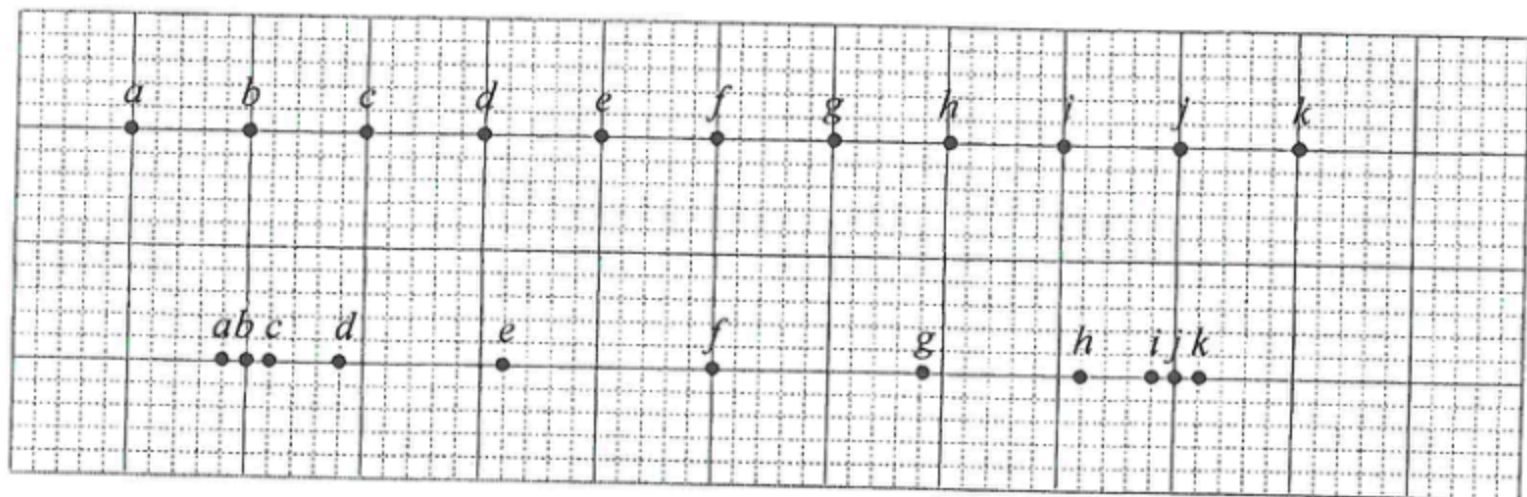


Figure (1)

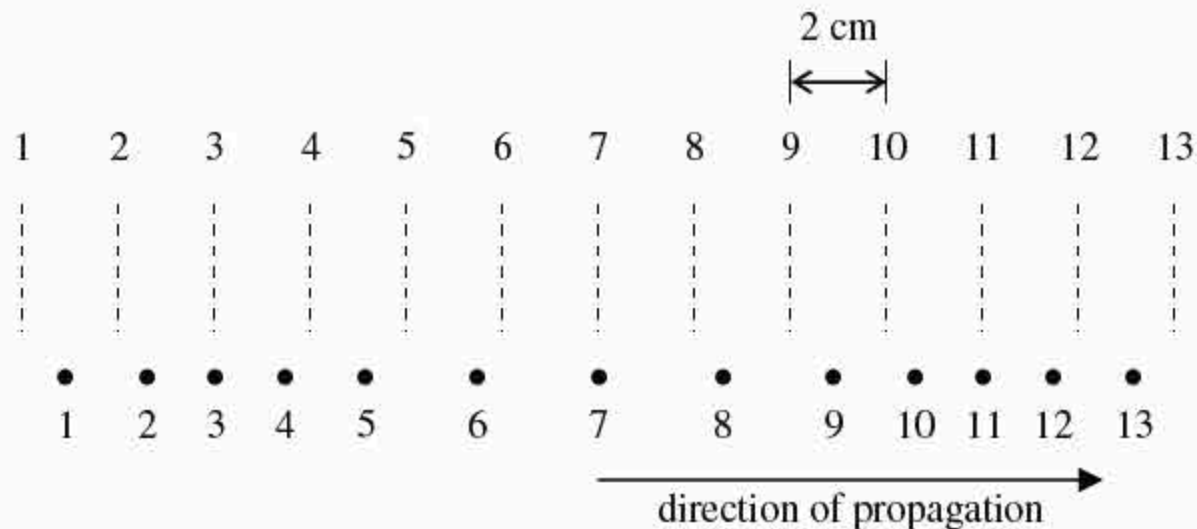
Figure (2)

Figure (1) shows the equilibrium positions of particles *a* to *k* in a medium. A longitudinal wave is travelling through the medium and the positions of the particles at a certain instant are shown in Figure (2). Which of the following statements is/are correct at that instant ?

- (1) The separation between *c* and *k* is one wavelength.
- (2) *f* is momentarily at rest.
- (3) *i* and *k* are moving in the same direction.

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

17.

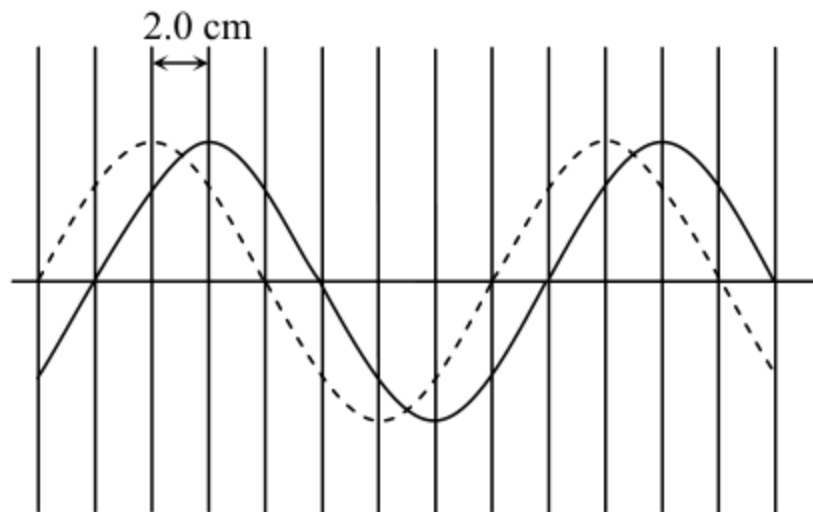


A longitudinal wave travels to the right through a medium containing a series of particles. The figure above shows the positions of the particles at a certain instant. The dotted lines indicate the equilibrium positions of the particles. Which of the following statements about the wave at the instant shown is/are correct ?

- (1) The wavelength of the longitudinal wave is 16 cm.
- (2) Particles 8 and 10 are moving in the same direction.
- (3) Particle 3 is momentarily at rest.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

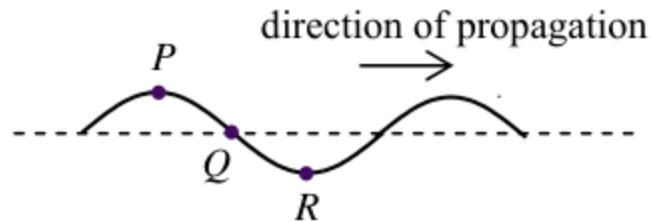
17.



The solid curve in the diagram shows a transverse wave at a certain instant. After 0.05 s, the wave has travelled a distance of 2.0 cm and is indicated by the dashed curve. Find the wavelength and frequency of the wave.

	Wavelength/cm	Frequency/Hz
A.	8	2.5
B.	16	2.5
C.	8	5
D.	16	5

18.



The figure shows the shape of a transverse wave travelling along a string at a certain instant. Which statement about the motion of the particles P , Q and R on the string at this instant is correct ?

- A. Particle P is moving downwards.
- B. Particle Q is stationary.
- C. Particle R attains its maximum acceleration.
- D. P and Q are in phase.